it, you’re not going to notice that your gravitational pull toward it has appreciably reduced.

Once you figure out how to make measurements on millimeter scales, though, you can test whether or not the decrease in gravity is what you expected from the standard equations. Going back to the ink-on-paper analogy, if you drop a gallon of ink on a sheet of paper, it’ll still look like you have a gallon of ink. But if you measure it by the drop, you’ll notice losing some when it soaks into the paper’s fibers. If extra dimensions are the width of millimeters, and you can measure gravity changing even on those kinds of scales, then the amount of gravity you’re losing to that extra dimensional bulk becomes comparable to the amount you’re trying to detect. You’ll see a drop-off of gravitational strength that’s more rapid than what you’d predict from general relativity in a nonleaky space, and that will make it obvious that something isn’t right.

So far, while we still don’t have any alternative consensus on an explanation for gravity’s weakness, we also haven’t found any solid indication that this leakage is actually happening, despite getting better and better at measuring gravity at very small scales. As appealing as extra dimensions may sound from a theoretical point of view, their existence is still more in the category of intriguing possibility than a confirmed feature of our cosmos. And to a large degree, the original motivation for them has faded, as almost all the most compelling theories explaining gravity’s weakness through leakage have been ruled out, since they predict alterations at levels we should have already detected. Still, we’re continuing the search, because if extra dimensions do turn out to be real, they offer an entirely new view of gravity, and of the universe. Our entire universe being on a brane contained within a larger spacetime brings up the possibility that there are other universes out there, perhaps on nearby branes, which may be capable of influencing our own with their gravity. Even more dramatically, interactions between branes could provide a new scenario for the origin of our universe. And, ultimately, its destruction.

Enter: the ekpyrotic cosmos.

COSMIC APPLAUSE

The first time I encountered the ekpyrotic scenario for the origin (and fate) of the cosmos, it was at a very engaging physics lecture at Cambridge